

Where do emotion regulation and reward maps overlap?

Neurosynth + NeuroQuery | exploratory computational research poster | 28 September 2026

NOT PEER REVIEWED

Research question

How stable is the spatial overlap across thresholds, and does a second synthesis method show a related pattern?

Data and method

Three Neurosynth association maps: emotion regulation (247 studies), reward (922), and depression (502). We compared positive z maps at thresholds 2, 3, 4, and 5, then checked six-neighbor components against the AAL SPM12 atlas.

Primary result

At $z \geq 3$, emotion regulation and reward share **1,296 voxels** (Jaccard **0.067**). Jaccard decreases from 0.120 at $z \geq 2$ to 0.028 at $z \geq 5$. Twenty studies are assigned to both terms (8.1% of the emotion-regulation set).

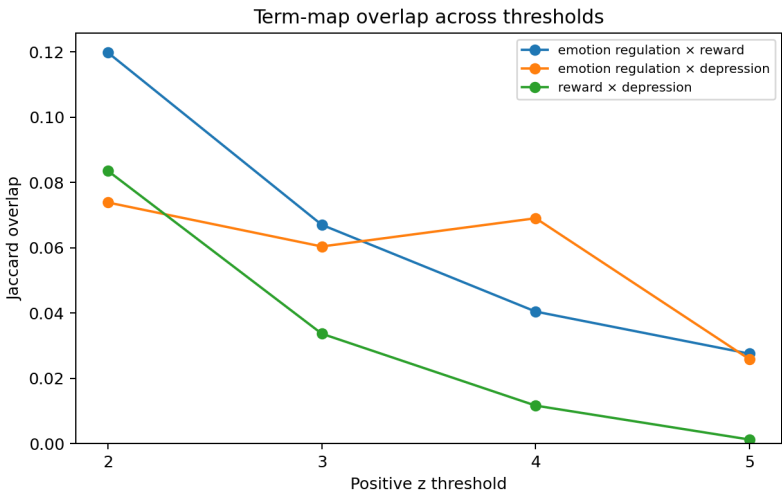


Figure 1. Descriptive threshold sensitivity across term pairs.

Where is the overlap?

The two largest $z \geq 3$ components contain 395 and 296 voxels. Their peaks are MNI (24, 2, -16) and (-18, 2, -14). AAL labels the first peak Amygdala_R; the second peak lies outside an AAL label, although much of its component is labeled Amygdala_L.

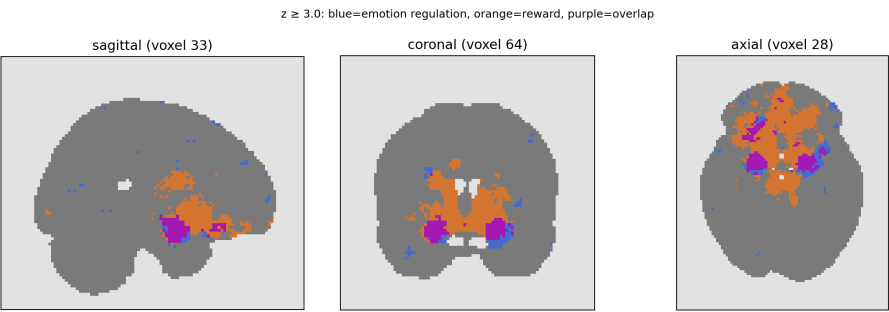


Figure 2. Blue: emotion regulation only. Orange: reward only. Purple: overlap. Gray silhouette: data footprint, not structural MRI.

Threshold stability

At $z \geq 5$, 88/395 (22.3%) and 76/296 (25.7%) of the original $z \geq 3$ component voxels remain. Smaller contiguous high-z cores persist; this is not a significance test.

Cross-method check

We generated NeuroQuery predicted maps for the same phrases and compared their spatial ranks with Neurosynth after resampling to a 4 mm grid. The top 1%, 5%, and 10% cutoffs are descriptive, not matched p-value thresholds.

Emotion regulation	rho 0.272 top-5% Jaccard 0.126
Reward	rho 0.502 top-5% Jaccard 0.391
Depression	rho 0.399 top-5% Jaccard 0.115

Interpretation

Reward has the highest focal cross-method agreement at all tested fractions. NeuroQuery predicts report locations; Neurosynth tests term-coordinate associations. Their literature overlaps, so this is methodological triangulation rather than independent replication.

Limits and next test

Keyword selection, publication and coordinate-reporting biases, atlas boundaries, study dependence, and spatial autocorrelation constrain interpretation. A follow-up should curate nonoverlapping studies and prespecify a coordinate meta-analysis and justified spatial null model.